WORKED SOLUTION

STEP III 1998, Question 3

Pure • Linear Recurrence Relations

STEP III 1998, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q3

The key idea

This is a first-order linear recurrence with a constant term, the discrete cousin of the differential equation $y' = cy - d$. The trick is the same as for the ODE: find the constant ("equilibrium") solution, then notice that the deviation from it simply gets multiplied by the same factor every day. For the limit, the only real work is understanding what $((1+c)^N - 1)/c$ does as $c \to 0$, and the binomial theorem settles it.

Solution

Solving for $c > 0$

We try $V_T = Ak^T + B$ for every T. Substituting into the recurrence,

$$Ak^{N+1} + B = (1+c)(Ak^N + B) - d \quad \Longleftrightarrow \quad Ak^N(k - (1+c)) = cB - d.$$

The right-hand side does not depend on N, while the left-hand side does unless it vanishes (we want $A \neq 0$ and $k \neq 0$, or the form is just a constant). So we need both sides to be zero for every N:

$$k = 1 + c, \qquad B = \frac{d}{c},$$

where dividing by c is fine because $c > 0$. With these values, $V_T = A(1+c)^T + d/c$ satisfies the recurrence for any constant A. Putting $T = 0$ gives $V_0 = A + d/c$, so $A = V_0 - d/c$ and

$$\boxed{V_N = \left(V_0 - \frac{d}{c}\right)(1+c)^N + \frac{d}{c}}$$

Ask yourself We guessed a form and it worked. How do we know there isn't some other solution?

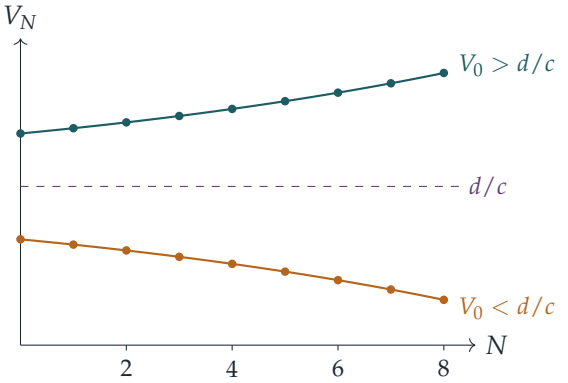
The recurrence determines V_1 from V_0, then V_2 from V_1, and so on, so there is exactly one sequence with a given V_0. Our formula satisfies the recurrence and has the right V_0, so it *is* that sequence. (If you prefer a direct argument, put $U_N = V_N - d/c$; the recurrence becomes $U_{N+1} = (1+c)U_N$, so $U_N = (1+c)^N U_0$, which is the same answer.)

It will be useful to rearrange the answer as

$$V_N = V_0(1+c)^N - d\,\frac{(1+c)^N - 1}{c}. \qquad (*)$$

This has a nice reading: the first term is V_0 growing at rate c per day, and the second is $d(1 + (1+c) + \cdots + (1+c)^{N-1})$, the daily deductions each grown for the days remaining, summed as a geometric series.

The constant d/c is the equilibrium value. If $V_0 > d/c$ the bond grows without bound, if $V_0 = d/c$ it stays put, and if $V_0 < d/c$ it falls away ever faster:



(The picture uses $c = 0.1$ and $d/c = 5$, with $V_0 = 6$ and $V_0 = 4$.)

The case $c = 0$

Now the recurrence is $V_{N+1} = V_N - d$: the value drops by d each day, an arithmetic progression. So

$$\boxed{V_N = V_0 - Nd}$$

Notice our $c > 0$ formula cannot simply be evaluated at $c = 0$, because of the d/c; that is exactly why the question asks about the limit.

The limit as $c \to 0$

Fix N and use $(*)$. The first term tends to V_0, since $(1+c)^N \to 1$ for fixed N. For the second, expand by the binomial theorem:

$$(1+c)^N - 1 = Nc + \binom{N}{2}c^2 + \cdots + c^N,$$
$$\text{so} \quad \frac{(1+c)^N - 1}{c} = N + \binom{N}{2}c + \cdots + c^{N-1}.$$

Because N is fixed, this is a polynomial in c with finitely many terms, and every term after the first contains a positive power of c. Hence it tends to N as $c \to 0$, and

$$\lim_{c \to 0} V_N = V_0 - Nd,$$

which is the $c = 0$ solution, as required. (Equivalently, $\left((1+c)^N - 1\right)/c$ is the difference quotient for x^N at $x = 1$, so its limit is the derivative N.)

The words "for fixed N" matter: with c small but N large, terms like $\binom{N}{2}c$ need not be small, and the exponential growth eventually wins over the linear behaviour.

Checks

- $N = 1$: the formula gives $(V_0 - d/c)(1+c) + d/c = (1+c)V_0 - d$, which is the recurrence itself.
- $V_0 = d/c$ gives $V_N = d/c$ for all N, the equilibrium.
- Numerically, with $V_0 = 7$, $d = 2$, $N = 12$: the formula gives $V_{12} \approx -17.48$ for $c = 0.01$ and -17.005 for $c = 10^{-4}$, approaching $V_0 - 12d = -17$. The closed form also agrees exactly with direct iteration in many random rational cases.

Common slips

- Taking $B = -d/c$: the sign comes from $cB - d = 0$, so $B = +d/c$.
- Using $A = V_0$ instead of $A = V_0 - d/c$: the constant B also contributes at $T = 0$.
- Trying to set $c = 0$ directly in the $c > 0$ formula, which is undefined; the limit is needed.

Beyond STEP

Recurrences like this are *linear difference equations*, and their theory runs parallel to linear ODEs: the general solution is a particular (here constant) solution plus the general solution of the homogeneous equation $U_{N+1} = (1+c)U_N$. In finance this is the formula for a loan or annuity with fixed repayments, and the $c \to 0$ limit is the passage from

compound to simple behaviour. Boyce and DiPrima, *Elementary Differential Equations*, includes a section on first-order difference equations alongside the ODE theory.

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